

Group 1: Anorexia Treatment Study

Dataset Guidance

Dataset Overview

File: anorexia.csv

Sample Size: $n = 72$ patients with anorexia nervosa (Cont=26, CBT=29, FT=17)

Research Context: Clinical trial comparing three treatments for anorexia: Control, Cognitive Behavioral Therapy (CBT), and Family Therapy (FT). Weight measured before and after treatment.

Variables

- **Treat:** Treatment group (Cont, CBT, FT)
- **Prewt:** Pre-treatment weight (pounds)
- **Postwt:** Post-treatment weight (pounds)

Loading the Data in R

```
# Load the dataset
anorexia <- read.csv("anorexia.csv")

# View structure
str(anorexia)
head(anorexia)

# Summary by treatment group
tapply(anorexia$Prewt, anorexia$Treat, summary)
tapply(anorexia$Postwt, anorexia$Treat, summary)
```

Research Question to Consider

“Which treatment is most effective for increasing weight in anorexia patients?”

Alternatively: “Do the three treatments differ in their effectiveness?”

Key Decision Point

The Challenge: You have PRE and POST measurements for each person. How will you handle this in your analysis using methods covered in this course?

Questions to explore with AI:

- How do statisticians typically handle pre/post measurements?
- Should you analyze change scores? Final weights? Both?
- What methods from this course can compare three groups?

Things to Think About

Exploratory Analysis:

- Create visualizations (boxplots, histograms) to understand the data
- Calculate summary statistics by treatment group
- Consider: What would you calculate for each person? (Change? Percent change?)

Statistical Methods:

- Review methods for comparing groups
- Consider: How many groups are you comparing?
- Consider: What would happen if you tested each treatment separately vs. all together?

Baseline Measurements:

- You have pre-treatment weights - how should you use this information?
- Different approaches: ignore baseline, calculate change, adjust for baseline
- Which approaches are possible with methods you've learned?

AI Consultation Strategy

Important: When you ask AI tools for analysis suggestions, they may recommend methods beyond this introductory course.

Prompts to try:

- “How should I analyze this anorexia treatment data with pre and post weights?”
- “What’s the difference between ANOVA on change scores vs. other approaches?”
- “I haven’t learned [method AI suggested]. Can I answer my question with simpler methods?”

Watch for:

- Tools suggesting different analytical approaches
- Methods you haven’t encountered in this course
- Disagreement between tools about the “best” method

Document everything: Save what each AI tool recommends - this is material for your AI comparison section!

Questions for Your Analysis

As you work through this dataset, consider:

1. What research question are you answering?
2. Which statistical method(s) from this course can address this question?
3. What assumptions do you need to check?
4. How will you interpret your results in the context of the research question?
5. If AI suggested advanced methods, why did you choose a simpler approach?

Resources

- Week 9 drop-in session (Wed 12-1:20pm, 153 Milam)
- Textbook: Chapter 12 (Analysis of Variance)
- Your team!
- Canvas discussion board

Remember: There may be multiple valid approaches to analyzing this data. The goal is to:

- Choose a method you understand from this course
- Justify why your approach answers the research question
- Critically evaluate what different AI tools suggested
- Explain your reasoning clearly in your report